what the sensor measures

digitalisierung und programmierung · input, session 3 agenda

1 measuring settles it

2 what the sensor really delivers

3 one reading is a guess

4 the knob

5 measuring like you mean it

measuring settles it

which one lands first?

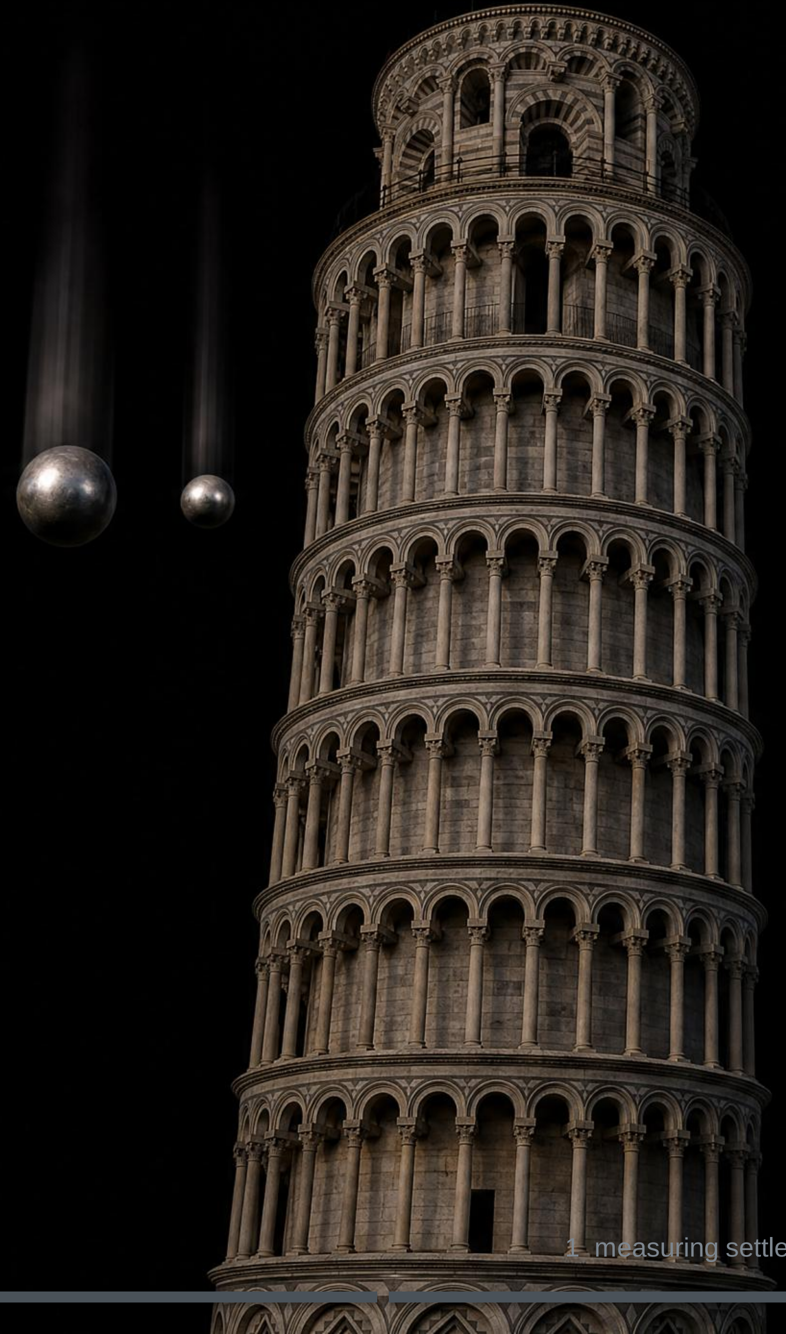


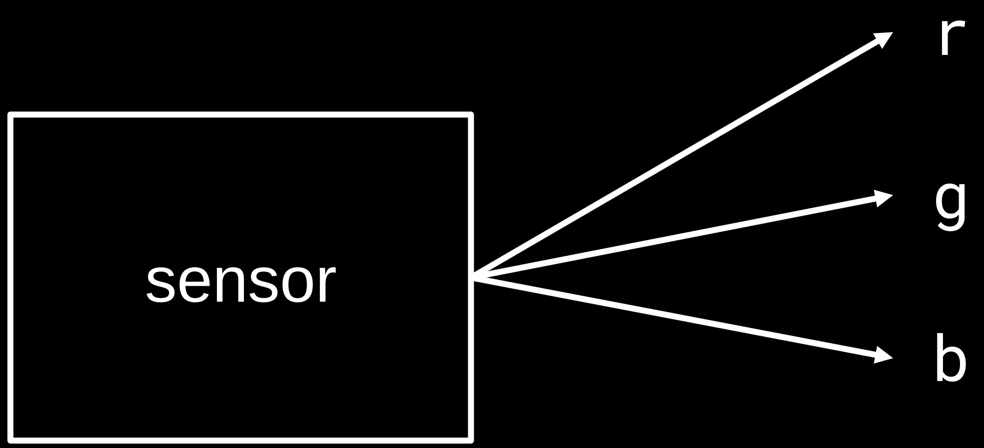
image: ai-generated (gpt-image-2)

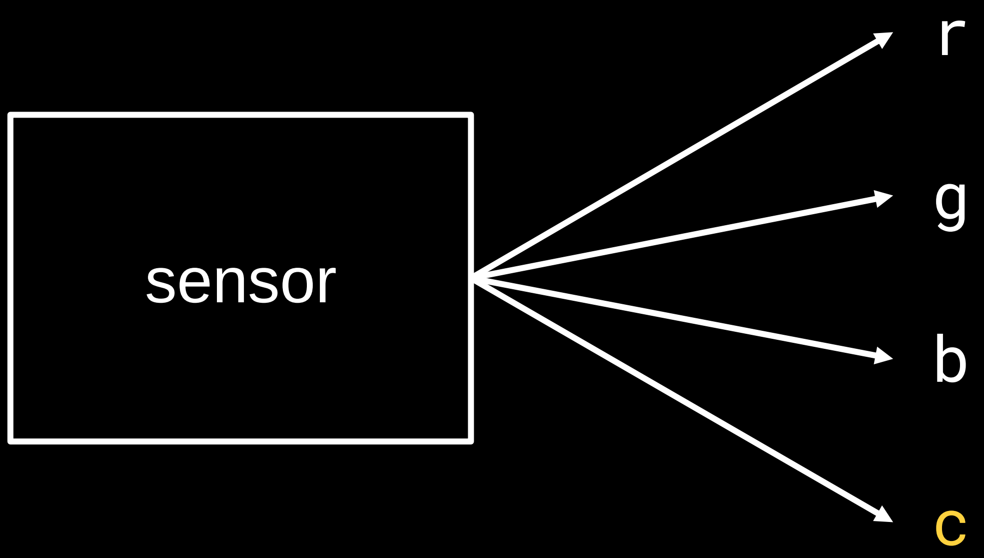
back to the box

light → `classify()` → "red"

today we live here.

what the sensor really delivers


sensor
r
g
b



unfiltered. sees everything.
more sensitive.

sent is not measured

	what you send	what arrives
r	255	203
g	0	41
b	0	57

not broken. normal.

calibrate, then classify

nearest match wins.

one reading is a guess

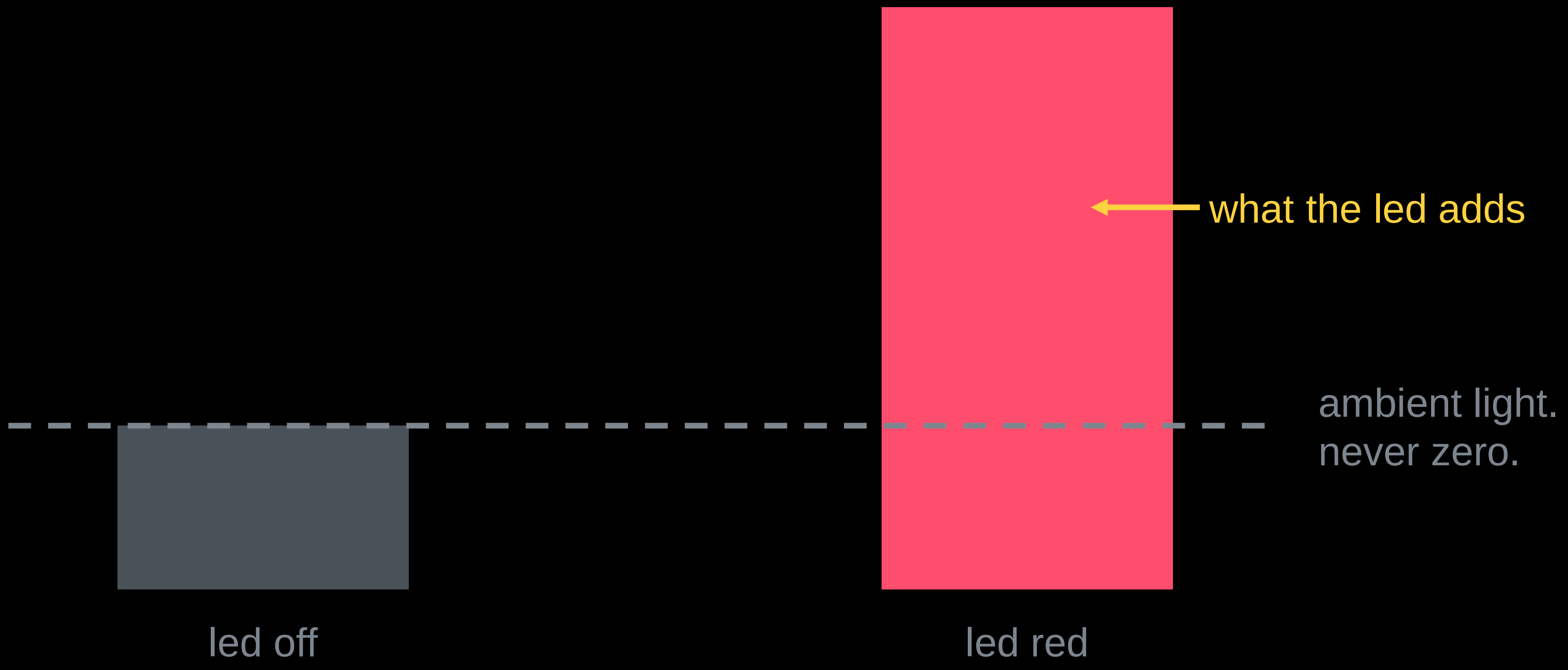
198   205   201   195   206

which one is true?

readings scatter.
average many.

mean of the five: about 201.

measure the dark



measure the dark first.

the knob

the knob: integration time

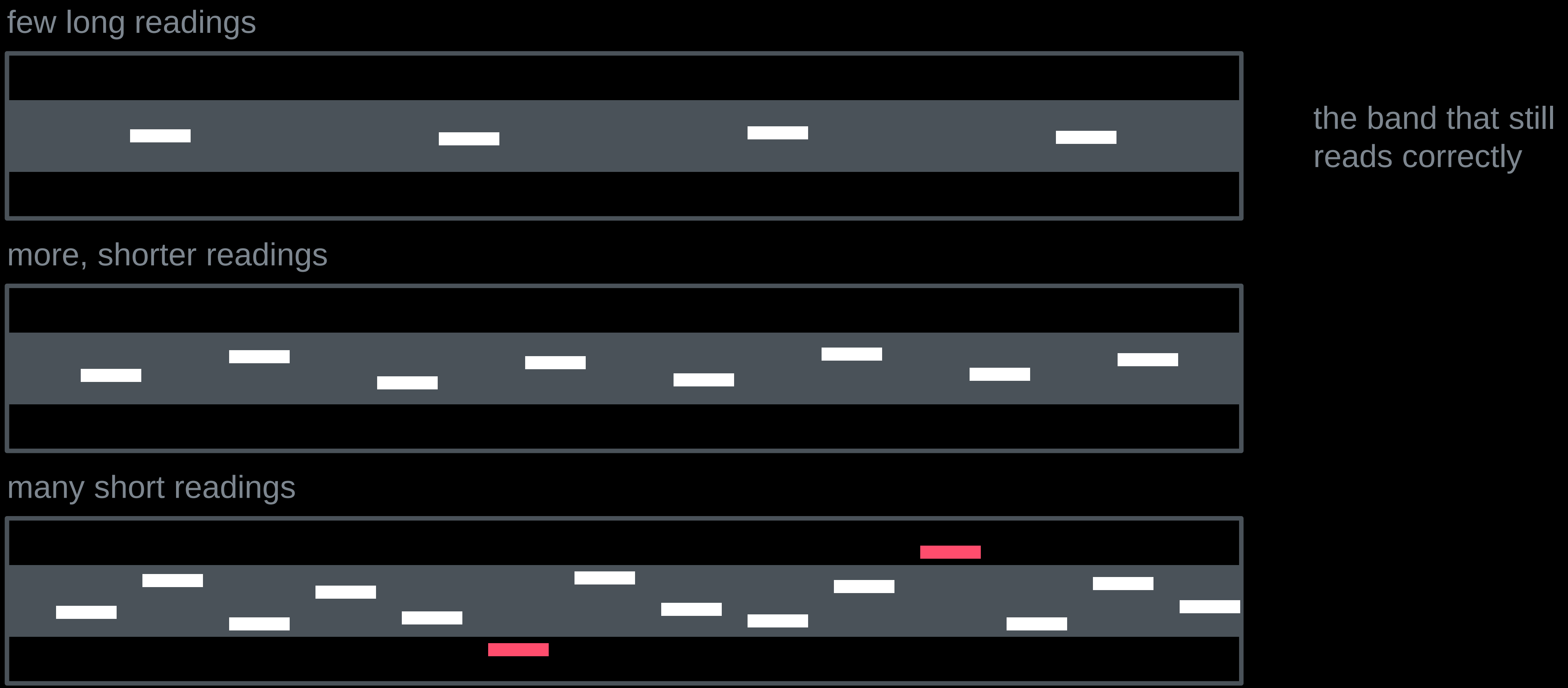
few long readings

the knob: integration time

few long readings

more, shorter readings

the knob: integration time



longer is calmer.
shorter is faster.

where is your limit? measure it.

measuring like you mean it

two teams, two ways to change things

team a

changed distance and
integration time.
result: better.

team b

changed only the distance.
result: better.

who learned something?

same conditions, written down

one variable changed

enough repetitions

expectation written down before

your measurement log is exactly this list, plus every value, including the inconvenient ones.

luck or skill?

5 in a row.
guessing does this
once every 32 tries.

50 in a row.
guessing never
does this.

the assistant knows sensors

"increase the gain, that
improves recognition."

your measurement knows yours

recognition rate
went down.

document it. that is the error log.

if you did not measure it,
you do not know it.